
A Drift-Robust E-Process for Nested Forecast Comparison

Anonymous

Abstract

Tests of out-of-sample predictive ability compare a model against a benchmark, and handle the outcome’s unconditional drift by plugging in an estimate of it. When the benchmark is the zero forecast, the resulting statistic loads on that drift: we show by simulation that Clark–West rejects a *pure-noise* signal 81% of the time at the nominal 5% level once the asset drifts at an annualised Sharpe ratio of 2. We give an e-process that removes the drift instead of estimating it, by taking an infimum over the nuisance of a convex combination of a signal-betting martingale and a drift-centring martingale. Ville’s inequality then bounds the rejection probability at α in finite samples, uniformly in time and uniformly in the unknown drift; measured, it never exceeds 1% across the same sweep. We report what the construction costs rather than only what it fixes: against a correctly specified Clark–West test it needs roughly twice the sample at matched size, and anytime-validity itself accounts for a factor of 1.90. Inverting the process over its value parameter yields a time-uniform *upper* bound on incremental predictive value, which is available whether or not anything is rejected. Applied to a century of equity-premium data it certifies nothing, and the median of its marginal 95% upper endpoints on incremental annualised information ratio is 0.60.

1 The failure this fixes

Context. E-values and test martingales [Ville, 1939, Shafer and Vovk, 2019, Grünwald et al., 2024, Ramdas et al., 2023, Waudby-Smith and Ramdas, 2024] give anytime-valid inference with merging rules [Vovk and Wang, 2021] and false-discovery control [Wang and Ramdas, 2022] that hold under arbitrary dependence. They are not yet used in the forecast-evaluation literature, where the standard toolkit remains fixed-sample. This note supplies the missing construction for one specific and consequential composite null, and reports its price.

Let y_t be an outcome and g_t a predictable signal. A forecast-comparison test asks whether g_t carries information about y_t beyond a benchmark. Under nesting the Diebold–Mariano statistic is invalid, because the loss differential is degenerate under the null [Clark and McCracken, 2001]; Clark–West [Clark and West, 2007] supplies the standard adjustment and is what the applied literature uses.

Both handle the outcome’s unconditional mean by estimating it. That is the weak point. Table 1 sweeps the drift of an asset whose returns are *independent of the signal*, so every rejection is a false one. Against a zero benchmark, Clark–West’s rejection rate rises with the drift and reaches 0.81; the statistic is rewarding the asset for having gone up. Against an estimated-mean benchmark it is calibrated, but that repair costs the estimate’s own error and still gives no guarantee at a data-dependent stopping time.

Table 1: False rejections at nominal 5% against a signal that is pure noise, as the asset’s drift grows. 400 replications per row, 800 daily observations. The certificate’s bound holds uniformly in the drift, which is the property being illustrated. Source: `certificate_drift_sweep.csv`.

Asset’s annualised Sharpe	Clark–West vs. zero	Clark–West vs. mean	E-process
0.0	0.053	0.058	0.003
0.4	0.090	0.063	0.003
0.8	0.118	0.058	0.003
1.2	0.330	0.063	0.003
1.6	0.560	0.055	0.010
2.0	0.810	0.040	0.008

2 Construction

Let $\mathcal{F}_t$ be the filtration generated by everything observable through period t , and let g_t be $\mathcal{F}_{t-1}$ -measurable. The null is composite in the drift,

$$H_0 : \mathbb{E}[y_t - \mu | \mathcal{F}_{t-1}] = 0 \quad \text{for some fixed but unknown } \mu \in \mathbb{R},$$

so the signal carries no information about the outcome, whatever the outcome’s mean.

Fix a candidate μ and put $\psi_t(\mu) = (y_t - \mu)/\hat{\sigma}_t$, with $\hat{\sigma}_t$ the running scale through $t - 1$. Define two non-negative martingales,

$$\begin{aligned} W_t^{\text{sig}}(\mu) &= \prod_{s \leq t} (1 + \lambda_s g_s \psi_s(\mu)), & \lambda_s &\in [0, c/(|g_s|E)] \text{ predictable,} \\ W_t^{\text{ctr}}(\mu) &= \frac{1}{2} \prod_{s \leq t} (1 + \kappa \psi_s(\mu)) + \frac{1}{2} \prod_{s \leq t} (1 - \kappa \psi_s(\mu)), & \kappa &= c'/E, \end{aligned}$$

where E is an a-priori envelope on $|\psi|$ and $c, c' < 1$ keep every factor positive. The first is the wealth of a bettor staking on the signal leading the outcome; the second is the average of two bettors staking in opposite directions on μ being the wrong centre.

Proposition 1 (Validity). *With $w \in (0, 1)$, let $\mathcal{E}_t = \inf_{\mu \in \mathbb{R}} [w W_t^{\text{sig}}(\mu) + (1 - w) W_t^{\text{ctr}}(\mu)]$. Under H_0 , $P(\exists t : \mathcal{E}_t \geq 1/\alpha) \leq \alpha$ in finite samples, uniformly in t and uniformly in the unknown drift.*

Proof sketch. At the true drift μ^* both factors are non-negative martingales with unit initial value, hence so is their convex combination. An infimum over μ is bounded above by the value at μ^* , so $\mathcal{E}$ is dominated by a non-negative martingale and Ville’s inequality applies. Predictability of λ_s is what makes each factor fair; no choice of λ can affect validity, only power. $\square$

The two terms play distinct roles. At a wrong μ the centring bettor gets rich, so the infimum is not dragged to zero by candidates the data have already ruled out; at the true μ the centring bettor makes nothing and the signal bettor drives the process. The nuisance is *eliminated* rather than estimated, and the elimination costs one line of proof rather than a bootstrap. The infimum is a standard device for composite nulls and is not claimed as new; it is also not growth-optimal, and the numeraire e-variable [Larsson et al., 2025] would be tighter. Computing that for this null is the obvious next step and we have not done it.

Remark 2 (The one numerical trap). *The infimum is computed on a grid, and the spacing must resolve the drift to better than its sampling error $\hat{\sigma}/\sqrt{n}$. A grid spaced at $\hat{\sigma}$ places the numerical infimum above the true one and leaks type-I error — we measured 0.274 at nominal 0.05 before diagnosing it. Our released implementation exposes a resolution helper and a regression test that reproduces the failure.*

3 What it costs

A construction that only reports its advantages is not useful for deciding whether to adopt it. Table 2 compares power against Clark–West where Clark–West is *correctly* specified, so the comparison is the unfavourable one. At matched empirical size the e-process needs roughly twice the sample to reach comparable power.

Table 2: Power at matched empirical size, 400 replications. “Learned” selects stakes from past payoffs; “pre-committed” fixes them from a design ratio. Clark–West wins wherever it is well specified. Source: `power_head_to_head.csv`.

True annualised IR	Clark–West	E-process (learned)	E-process (pre-committed)
1.0	0.210	0.143	0.163
1.5	0.495	0.335	0.390
2.0	0.850	0.710	0.758
3.0	1.000	0.983	0.998

Part of that gap is the price of anytime-validity itself and is not specific to this construction. Comparing the sample a test martingale needs to cross $1/\alpha$ with 80% probability against the fixed-sample size for 80% power at the same α gives a ratio of **1.90**. We state this because we previously stated it wrongly: an earlier draft compared a *median* crossing time with an 80%-power sample size and reported a 3% cost. The corrected figure is a factor of two, and the correction is in the release history.

4 A ceiling, not only a verdict

Inverting the same construction over a value parameter rather than testing it against zero gives a time-uniform confidence interval for $\theta = \mathbb{E}[z_t(y_t - \mu)]$, the expected return of a position sized by the standardised signal. Its upper endpoint is a bound on incremental predictive value that is available *whether or not anything is rejected* — which is what a failed test cannot otherwise supply, since a non-rejection is consistent both with no effect and with an effect the design could never have seen.

Applied to 117 predictor–frequency pairs from a century of equity-premium data [Goyal et al., 2024], nothing is certified under any of nine specifications, and neither Clark–West, the Clark–McCracken encompassing test, nor a model confidence set [Hansen et al., 2011] rejects on the same forecasts. The interval is the informative output: across the 116 unflagged pairs the median marginal 95% upper endpoint on the incremental annualised information ratio is 0.60, and 112 exceed the 0.25 that this literature treats as economically meaningful. A century of the most studied data in finance separates neither hypothesis.

5 Limitations

The e-process is less powerful than Clark–West wherever Clark–West is well specified, and we recommend it only where the drift is genuinely a nuisance, where monitoring is sequential, or where a bound rather than a verdict is wanted. Validity requires an a-priori envelope on the payoff, which must be asserted rather than fitted; a fitted envelope makes the stakes depend on the future. The identity payoff assumes only a martingale-difference null, but bounded payoffs such as $\tanh$ additionally require conditional symmetry, which we tested and rejected on our own data (skew -1.05 , $p = 2.7 \times 10^{-78}$) — hence identity is the default.

Software. The construction, the ceiling, e-value merging and e-BH, and the design calculations are released as a documented package with ~ 750 lines of tests, depending only on `numpy` and `scipy`. Calibration and power tables in this paper are reproduced by one script each. (*Repository link withheld for double-blind review.*)

References

- Jean Ville. Étude critique de la notion de collectif. *Monographies des Probabilités*, 1939.
- Glenn Shafer and Vladimir Vovk. *Game-Theoretic Foundations for Probability and Finance*. Wiley, Hoboken, NJ, 2019.
- Peter Grünwald, Rianne de Heide, and Wouter M. Koolen. Safe testing. *Journal of the Royal Statistical Society Series B*, 86(5):1091–1128, 2024.
- Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023.

- Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B*, 86(1):1–27, 2024.
- Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021.
- Ruodu Wang and Aaditya Ramdas. False discovery rate control with e-values. *Journal of the Royal Statistical Society Series B*, 84(3):822–852, 2022.
- Todd E. Clark and Michael W. McCracken. Tests of equal forecast accuracy and encompassing for nested models. *Journal of Econometrics*, 105(1):85–110, 2001.
- Todd E. Clark and Kenneth D. West. Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1):291–311, 2007.
- Martin Larsson, Aaditya Ramdas, and Johannes Ruf. The numeraire e-variable and reverse information projection. *Annals of Statistics*, 53(3), 2025.
- Amit Goyal, Ivo Welch, and Athanasse Zafirov. A comprehensive 2022 look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 37(11):3490–3557, 2024.
- Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.